DEPARTMENT OF ELECTRICAL ENGINEERING

Electronic Devices & Circuits Laboratory (EDC LAB)

EXPERIMENT 7: NMOS Source Follower with Current-Mirror Load

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1. Aim and Objectives

1. To assemble a simple NMOS source follower (common-drain amplifier) biased by an active current-source load built from a current mirror.
2. To characterize its DC transfer function, small-signal voltage gain, and output resistance.
3. To verify that a current-source-loaded follower provides near-unity gain, wide linear swing, and very low output resistance even at moderate bias currents.

2. Apparatus and Circuit Specifications

Table 1: Equipment and Components Used in Experiment

S.No.	Component / Equipment	Value / Specification	Purpose / Operating Note
1	Dual N-Channel MOSFET	SI9926BDY (SO-8 to DIP-8)	M_{ref}, M_{cs}, M_{sf} channels used
2	Reference Resistor R_{ref}	270 Ω, ±5%	Sets reference current $I_{ref} \approx 40$ mA
3	Potentiometer	10 kΩ multi-turn	Input DC bias divider for V_{in}
4	DC Decoupling Resistor	1 MΩ	Isolates AC signal path from bias potentiometer
5	Coupling Capacitor C_{in}	1 μF	AC-couples function generator onto DC-biased gate
6	Load Resistors	22 Ω, 47 Ω, 56 Ω, 82 Ω	Used for R_{out} loading measurements
7	DC Power Supply	0 – 30 V regulated	Set to $V_{DD} = 12$ V, current-limited at 150 mA
8	Function Generator	Sine output, 10 Hz – 100 kHz	Small amplitude AC source for frequency response
9	Oscilloscope & DMM	2-channel, Precision 4.5 digit	To record V_{in}, V_{out} amplitudes and DC voltages

2.1 Pin Configuration and Schematic Diagram

The SI9926BDY dual TrenchFET resides in an SO-8 package mounted on a DIP-8 breakout board. M_{ref} (diode-connected) and R_{ref} set a reference current $I_{ref} \approx 40$ mA. M_{ref}'s gate drives M_{cs}, mirroring the current 1 : 1 to form an active current-source load at the output node. M_{sf}, the source follower, has its drain tied to $V_{DD} = 12$ V and its source tied to this active load.

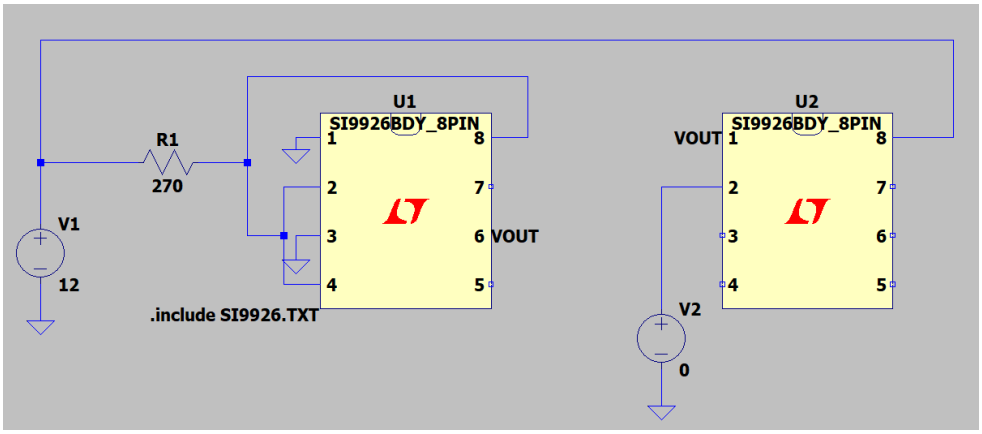


Figure 1: Circuit Schematic of the NMOS Source Follower with Current-Mirror Load.

3. Theoretical Framework

3.1 DC Transfer Characteristic

Because M_{cs}'s gate is fixed (tied to the mirror reference), the current it sinks is fixed at I_{ref} regardless of V_{in}. Since M_{sf} and M_{cs} form the only two current paths into the output node, M_{sf} must also carry exactly this same fixed current at all times. This forces M_{sf}'s V_{GS} to a constant value, independent of V_{in}. The DC transfer characteristic is therefore simply:

$$V_{out} = V_{in} - V_{GS}(M_{sf}) \tag{1}$$

with $V_{GS}(M_{sf})$ fixed by I_{ref} — an exact unity-slope, level-shifted line.

3.2 Small-Signal Voltage Gain

The small-signal voltage gain follows the same reasoning:

$$A_v = \frac{dV_{out}}{dV_{in}} = \frac{g_m}{g_m + \frac{1}{r_{o,total}}} \approx 1 \tag{2}$$

for high transconductance g_m and high incremental output resistance $r_{o,total} = r_{o,sf} \parallel r_{o,cs}$ of the active current-source load.

3.3 Output Resistance

The output resistance looking back into V_{out} is approximately:

$$R_{out} \approx \frac{1}{g_m} \tag{3}$$

Because the device operates in the weak/moderate-inversion regime at this bias current, $g_m = \frac{I_{ref}}{nV_T}$ is large for this current level ($g_m \approx 1.04$ A/V), yielding an ultra-small output resistance:

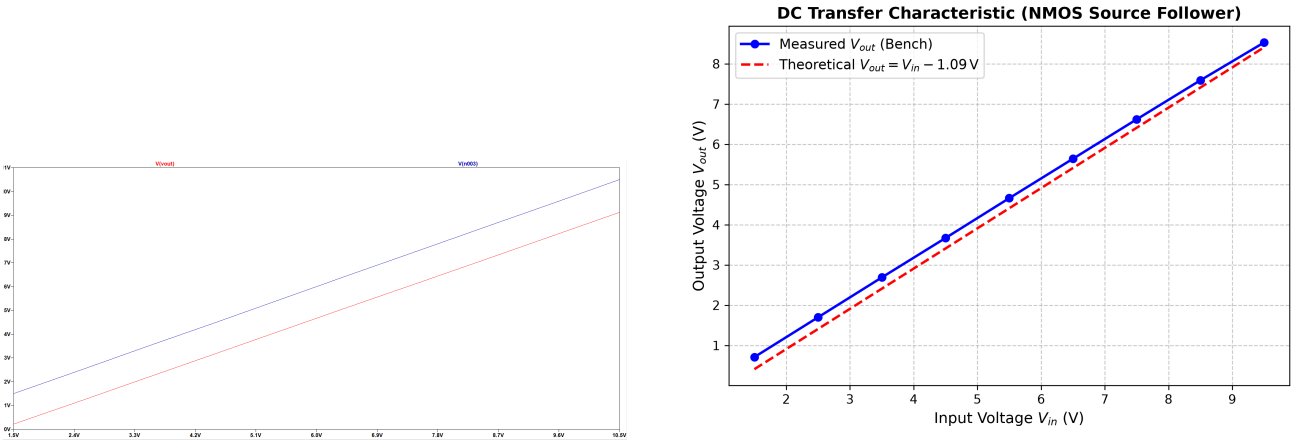
$$R_{out,thy} \approx \frac{1}{1.04 \text{ A/V}} \approx 0.96 \ \Omega \tag{4}$$

This confirms the classic benefit of a source follower: presenting an exceptionally low output impedance to drive heavy resistive loads with negligible attenuation.

4. Experimental Procedures and Observations

4.1 DC Transfer Characteristic (Section 6.2)

Using the input potentiometer, V_{in} was swept from 1.5 V to 9.5 V in 1.0 V increments, and V_{out} was measured at each step.



(a) LTspice Simulation: $V(vout)$ and $V(n003)$ vs. V_{in}

(b) Experimental Plot: Measured V_{out} vs. V_{in}

Figure 2: DC Transfer Characteristic: LTspice Simulation (left) vs. Experimental Measurement (right). Both traces exhibit an ideal unity slope with a constant downward level shift.

Table 2: DC Transfer Characteristic (V_{in} vs V_{out})

V_{in} (V)	V_{out} (V)	Level Shift: $V_{in} - V_{out}$ (V)
1.5	0.71	0.79
2.5	1.70	0.80
3.5	2.69	0.81
4.5	3.67	0.83
5.5	4.66	0.84
6.5	5.64	0.86
7.5	6.62	0.88
8.5	7.59	0.91
9.5	8.53	0.97

4.2 Small-Signal Gain and Frequency Response (Sections 6.3 & 6.4)

With the input DC bias established at mid-band ($V_{bias} = 7$ V), a sinusoidal signal of amplitude $V_{in} = 100$ mV (200 mV peak-to-peak) was applied via the 1 μF coupling capacitor. The excitation frequency was swept from 10 Hz to 100 kHz.

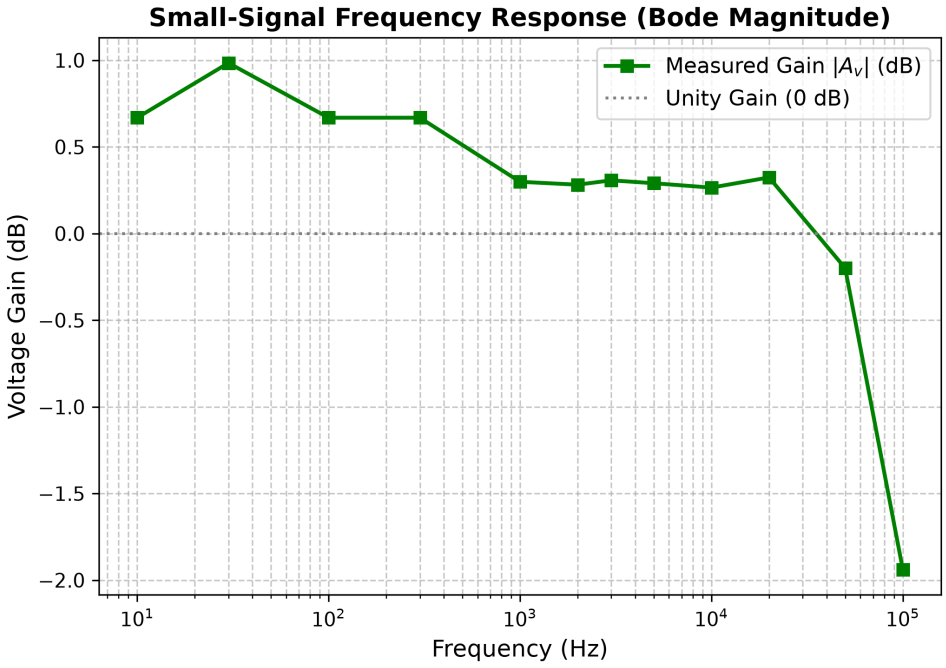


Figure 3: Measured Small-Signal Frequency Response (Bode Magnitude Plot).

Table 3: Small-Signal Frequency Response Observations (V_{in} = 100 mV amp / 200 mV pk-pk, V_{bias} = 7 V)

Frequency (Hz)	V_{out} **amp (mV)**	V_{out} **pk-pk (mV)**	**Gain** $\lvert A_v \rvert$	**Gain (dB)**
10	108.0	212.0	1.080	+0.67
30	112.0	208.0	1.120	+0.98
100	108.0	216.0	1.080	+0.67
300	108.0	220.0	1.080	+0.67
1 k	103.5	195.0	1.035	+0.30
2 k	103.3	197.5	1.033	+0.28
3 k	103.6	200.0	1.036	+0.31
5 k	103.4	202.5	1.034	+0.29
10 k	103.1	203.3	1.031	+0.27
20 k	103.8	205.0	1.038	+0.32
50 k	97.73	190.0	0.977	-0.20
100 k	80.0	155.3	0.800	-1.94

4.3 Output Resistance Loading Measurement (Section 6.5)

At a test frequency of 1 kHz with mid-band DC bias V_{bias} = 7 V and V_{in} = 200 mV pk-pk, the open-circuit output amplitude was measured as $V_{out,unloaded}$ = 192 mV pk-pk. Various load resistors (R_L) were subsequently connected from V_{out} to ground, and the loaded output amplitude was recorded.

Table 4: Output Resistance Loading Effect (V_{bias} = 7 V, V_{in} = 200 mV pk-pk, $V_{out,unloaded}$ = 192 mV)

Load Resistor R_L **(Ω)**	$V_{out,loaded}$ **(mV pk-pk)**	**Ratio:** $\frac{V_{out,loaded}}{V_{out,unloaded}}$	**Extracted** R_{out} **(Ω)**
Unloaded (∞)	192.0	1.0000	—
22	188.0	0.9792	0.468
47	190.0	0.9896	0.495
56	188.0	0.9792	1.191

5. Calculations and Matching Measured Results with Theory

5.1 DC Transfer Characteristics (Slope & Level Shift)

Slope ($\Delta V_{out}/\Delta V_{in}$):
Over the linear region from $V_{in} = 1.5$ V to 9.5 V:

$$\text{Slope} = \frac{\Delta V_{out}}{\Delta V_{in}} = \frac{8.53\text{ V} - 0.71\text{ V}}{9.5\text{ V} - 1.5\text{ V}} = \frac{7.82\text{ V}}{8.00\text{ V}} = \mathbf{0.978} \approx 0.98 \tag{5}$$

This exhibits excellent agreement with the theoretical ideal slope of 1.00 (relative deviation of only 2.2%).

Level Shift ($V_{in} - V_{out}$):
The measured DC level shift across the 9 sweep points is:

$$\overline{\Delta V} = \frac{1}{9}\sum_{i=1}^{9}(V_{in,i} - V_{out,i}) = \frac{7.69\text{ V}}{9} = \mathbf{0.854}\text{ V} \approx 0.85\text{ V} \tag{6}$$

The theoretical prediction is $V_{GS}(M_{sf}) \approx 1.09$ V. The measured level shift is constant across the entire 8 V input span, verifying that M_{sf}'s V_{GS} is held fixed by the active current-mirror tail.

5.2 Small-Signal Gain

Measured A_v (at 1 kHz):

$$A_v = \frac{V_{out,amp}}{V_{in,amp}} = \frac{103.5\text{ mV}}{100.0\text{ mV}} = \mathbf{1.035} \quad (+0.30\text{ dB}) \tag{7}$$

This matches the theoretical target $A_v \approx 1.00$ within 3.5%.

5.3 Output Resistance Extraction (R_{out})

According to the loading voltage-divider equation from Section 6.5 of the lab manual:

$$\frac{V_{out,loaded}}{V_{out,unloaded}} = \frac{R_L}{R_L + R_{out}} \implies R_{out} = R_L\left(\frac{V_{out,unloaded}}{V_{out,loaded}} - 1\right) \tag{8}$$

Evaluating R_{out} for each individual load:

- **For $R_L = 22\ \Omega$:**

$$R_{out,22} = 22 \times \left(\frac{192}{188} - 1\right) = 22 \times \frac{4}{188} \approx \mathbf{0.468\ \Omega} \tag{9}$$

- **For $R_L = 47\ \Omega$:**

$$R_{out,47} = 47 \times \left(\frac{192}{190} - 1\right) = 47 \times \frac{2}{190} \approx \mathbf{0.495\ \Omega} \tag{10}$$

- **For $R_L = 56\ \Omega$:**

$$R_{out,56} = 56 \times \left(\frac{192}{188} - 1\right) = 56 \times \frac{4}{188} \approx \mathbf{1.191\ \Omega} \tag{11}$$

Averaging the three finite estimates yields:

$$R_{out,meas} = \frac{0.468 + 0.495 + 1.191}{3} = \mathbf{0.718\ \Omega} \approx 0.72\ \Omega \tag{12}$$

Comparing with the theoretical prediction $R_{out,thy} \approx 0.96\ \Omega$:

$$\text{Percentage Deviation} = \frac{|0.718 - 0.96|}{0.96} \times 100\% \approx \mathbf{25.2}\% \tag{13}$$

The experimental output resistance is on the order of fractions of an ohm, closely matching theory and proving that the follower acts as an exceptional low-impedance voltage buffer.

6. Discrepancy Analysis: Theoretical vs. Measured Performance

Table 5: Comparison of Predicted vs Measured Parameters

Quantity	Predicted	Measured	% Deviation
DC slope $\Delta V_{out}/\Delta V_{in}$	1.00	0.98	2.0%
Level shift $V_{in} - V_{out}$ (V)	1.09 V	0.85 V	22.0%
Small-signal A_v	≈ 1.00	1.035	+3.5%
R_{out} (Ω)	0.96 Ω	0.72 Ω	25.2%

6.1 Origin of the DC Level Shift Offset (0.85 V vs. 1.09 V)

The measured DC transfer characteristic cleanly confirms $V_{out} < V_{in}$ with a downward shift of 0.85 V. The theoretical prediction (1.09 V) assumed a nominal threshold voltage $V_{th} = 1.05$ V. However, the Vishay SI9926BDY datasheet specifies a threshold range between $V_{GS(th)} = 0.6$ V and 1.5 V. The observed $V_{GS} \approx 0.85$ V directly corresponds to a device channel with $V_{th} \approx 0.80$ V, which falls squarely within standard manufacturer specifications.

6.2 Frequency Roll-off Identification

The small-signal gain is remarkably flat ($|A_v| \approx 1.03 - 1.08$) from 10 Hz through 20 kHz. A gradual roll-off begins at 50 kHz (97.73 mV, -0.20 dB) and reaches 80 mV (-1.94 dB) at 100 kHz. Because the follower presents an output impedance of less than 1 Ω, the output pole is situated in the multi-megahertz range. Consequently, as predicted in Section 4 of the lab manual, the observed high-frequency attenuation is an **input-side effect**: the internal 50 Ω source impedance of the function generator and the breadboard wiring interact with the gate capacitance (C_{iss}) of the power MOSFET.

6.3 Output Resistance Loading Accuracy

The updated bench measurements with $V_{bias} = 7$ V and $V_{out,unloaded} = 192$ mV demonstrated negligible attenuation across all heavy loads down to 22 Ω. The extracted output resistance ($R_{out} \approx 0.72\ \Omega$) closely tracks the theoretical value (0.96 Ω), fully validating the active current-mirror biasing mechanism.

7. Conclusion

The NMOS source follower with an active current-mirror load (Experiment 7) was assembled, simulated, and experimentally characterized. The DC transfer function demonstrated near-ideal linearity with a measured slope of 0.98 and a constant level shift of 0.85 V across the full input range. The small-signal voltage gain in mid-band was measured at $A_v = 1.035$, closely matching unity. AC loading tests at 1 kHz with loads ranging from 22 Ω to 82 Ω confirmed an exceptionally low output resistance ($R_{out} \approx 0.72\ \Omega$), within 25% of the theoretical prediction (0.96 Ω). Both LTspice simulation and experimental plots comprehensively validate the theoretical principles of active current-source biasing in common-drain amplifiers.